Medium-Term Outlook and Debt Sustainability

43. This section assesses the sustainability of Grenada's domestic and external public debt under the program and alternative scenarios.

44. **Under the program scenario, public debt is reduced from 120 percent of GDP in 2005 to close to 60 percent of GDP in 2015** (Table A1a).[15] A large part of this adjustment is achieved during the program period, so that by end-2008 public debt will amount to less than 95 percent of GDP. This sharp decline is largely a function of the sharp improvement in the primary balance during 2006–08 and a pick-up in GDP growth.

45. **In net present value (NPV) terms, public debt is at lower levels and also set to decline more modestly.** In NPV terms, total public debt falls from above 85 percent of GDP in 2005 to about 70 percent by end-2008 and to just above 50 percent by 2015. The NPV calculation takes into account the effects of the restructuring of Grenada's commercial debt, which implied a haircut of 40–45 percent on the NPV of future debt servicing flows (for exit yields in the 9–10 percent range), while leaving the face value of debt unchanged.[16]

46. **In addition to the recent commercial debt restructuring, rescheduling by official creditors is expected to lower debt servicing**. In particular, the program scenario assumes that eligible commercial creditors that did not participate in the recent debt restructuring will settle their claims in 2006 on the same terms as the recent debt exchange. In addition, it is assumed that Paris Club debt will be rescheduled on the basis of Houston terms and that the bilateral debt to Taiwan Province of China is restructured on terms comparable to those of the commercial debt exchange.

47. **Alternative scenarios underline the importance of the envisaged fiscal adjustment and measures to foster growth** (Table A1b, Figure 1). Keeping the primary deficit at its ten-year historical average of 2 percent of GDP, while also assuming GDP growth in line with the historical average (Alternative Scenario A1) would cause the NPV of government debt to rise to over 100 percent of GDP by 2015. With the program's fiscal adjustment achieved, but GDP growth rates permanently lower than assumed under the program scenario (Alternative Scenario A2), the NPV of debt would initially fall only to rise rapidly after 2010. Bound tests indicate considerable sensitivity to external shocks—in particular with respect to lower GDP growth in 2006/2007, which could bring the NPV of debt-to-GDP ratio on an unsustainable trajectory.

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[15] At the end of 2005, public debt amounted to 128 percent of GDP. However, total debt comprises contingent liabilities in the form of government-guaranteed debt, amounting to about 13 percent of GDP. It is debatable whether permanent fiscal adjustment is needed for all of the contingent liabilities. Against this background, it is assumed for the purpose of this exercise that only about half of these guarantees will be called. This assumption implies a starting debt-to-GDP ratio of 120 percent of GDP in 2005.

[16] See Box 2 in the main text of this report.

48. **The program scenario would allow for a considerable reduction of external vulnerability** (Table A2a). With the fiscal adjustment envisaged under the program, external public debt would fall from almost 85 percent of GDP at end-2005 to below 50 percent of GDP by 2015. In NPV terms, it would fall from almost 50 percent of GDP to below 40 percent over the same period. Bound tests suggest significant sensitivity to short-term shocks, but these effects peter out with time (Table A2b, Figure 2).

49. **All told, the adjustment envisaged under the program would bring public debt on a sustainable path**. As shown above, however, accomplishment of both fiscal and growth objectives are crucial to obtain this result. Failure to adhere to program targets would rapidly result in an unsustainable debt trajectory. And, unfortunately, a significant external shock, such as another major hurricane, could have a similar adverse effect.

Table A1a. Grenada: Public Sector Debt Sustainability Framework, Program Scenario, 2002–2025

(In percent of GDP, unless otherwise indicated)

	Actual				Estimate	Projections								
	2003	2004	Historical Average 5/	Standard Deviation 5/	2005	2006	2007	2008	2009	2010	2005-10 Average	2015	2025	2011-25 Average
Public sector debt 1/	110.0	129.5			120.1	111.7	103.8	94.3	87.8	81.5		60.8	33.3	
Of which: foreign-currency denominated	80.2	95.1			84.3	81.3	77.4	73.2	69.1	65.2		48.7	26.5	
Change in public sector debt	0.4	19.5			-9.4	-8.4	-7.8	-9.6	-6.5	-6.3		-3.5	-3.0	
Identified debt-creating flows	-5.3	3.5			-6.4	-10.3	-7.3	-8.9	-6.5	-6.3		-3.5	-3.0	
Primary deficit	-3.5	-5.5	2.0	4.7	-3.3	-1.1	-2.8	-5.5	-4.0	-3.8	-3.4	-2.7	-2.6	-2.6
Revenue and grants	39.5	42.9			49.1	47.1	43.5	44.9	43.5	43.4		42.3	42.3	
Of which: grants	6.9	7.5			11.0	8.0	2.7	2.5	1.2	1.1		0.0	0.0	
Primary (noninterest) expenditure	36.0	37.4			45.8	46.0	40.8	39.3	39.5	39.6		39.6	39.7	
Automatic debt dynamics	-1.8	9.0			-3.1	-9.3	-4.6	-3.4	-2.5	-2.5		-0.8	-0.3	
Contribution from interest rate/growth differential	-1.1	9.0			-2.2	-7.3	-4.6	-3.4	-2.5	-2.5		-0.8	-0.3	
Of which: contribution from average real interest rate	4.9	5.6			-0.3	0.0	0.8	0.6	1.1	0.9		1.6	1.1	
Of which: contribution from real GDP growth	-6.0	3.4			-1.9	-7.3	-5.3	-4.0	-3.6	-3.4		-2.5	-1.4	
Contribution from real exchange rate depreciation	-0.7	0.0			-0.9	-2.0	0.0	0.0	0.0	0.0		...	...	
Other identified debt-creating flows	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	5.7	16.0			-3.0	1.9	-0.5	-0.6	0.0	0.0		0.0	0.0	
NPV of public sector debt	...	84.6			85.9	80.9	76.6	70.3	67.5	64.2		51.5	28.9	
Of which: foreign-currency denominated	0.0	50.2			50.0	50.5	50.2	49.3	48.9	47.9		39.3	22.2	
NPV of contingent liabilities (not included in public sector debt)	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	8.2	16.9			7.1	8.7	6.0	3.0	4.9	4.2		4.9	5.7	
NPV of public sector debt-to-revenue ratio (in percent) 3/	...	197.1			174.8	172.0	176.1	156.7	155.3	147.9		121.7	68.3	
Of which : external	...	43.3			37.7	39.8	42.7	40.7	41.6	40.9		34.4	19.4	
Debt service-to-revenue ratio (in percent) 3/ 4/	12.3	29.1			10.1	10.1	9.5	9.2	11.0	9.5		11.1	16.0	
Primary deficit that stabilizes the debt-to-GDP ratio	-3.9	-25.0			6.1	7.4	5.1	4.1	2.5	2.5		0.8	0.3	
Key macroeconomic and fiscal assumptions														
Real GDP growth (in percent)	5.8	-3.0	3.2	4.3	1.5	6.5	5.0	4.0	4.0	4.0	4.2	4.0	4.0	4.0
Average nominal interest rate on forex debt (in percent)	6.7	5.7	6.4	2.5	1.6	2.6	2.4	2.3	2.9	2.9	2.5	4.5	5.0	5.0
Average real interest rate on domestic currency debt (in percent)	5.1	10.2	5.9	3.4	1.6	-1.1	1.5	1.3	2.2	2.0	1.2	3.4	3.6	3.7
Real exchange rate depreciation (in percent, + indicates depreciation)	-0.9	0.0	-0.2	1.3	-1.0	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	2.9	2.7	2.2	1.3	3.5	4.6	2.0	2.0	2.0	2.0	2.7	2.0	2.0	2.0
Growth of real primary spending (deflated by GDP deflator, in percent)	-15.8	-0.5	6.3	13.6	28.4	5.1	-6.5	1.0	4.4	4.2	6.1	4.0	4.0	4.0
Grant element of new external borrowing (in percent)	...	...	...	... ...	...	...	...	...	...	...	...	...	...	...

Sources: Country authorities; and Fund staff estimates and projections.

1/ Includes central government and central government guaranteed debt as well as debt of public enterprises. In 2005, part of the government guaranteed debt is excluded from the debt stock (see explanation in the text of the DSA annex)

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues including grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table A1b. Grenada: Sensitivity Analysis for Key Indicators of Public Debt 2005-2025

	Est.	Projections						
	2005	2006	2007	2008	2009	2010	2015	2025
NPV of Debt-to-GDP Ratio								
Program Scenario	86	81	77	70	68	64	51	29
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	85	86	87	87	90	92	103	129
A2. Permanently lower GDP growth 1/	86	82	80	75	75	74	80	124
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	86	90	94	91	93	94	103	126
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	86	88	91	84	82	80	70	53
B3. Combination of B1-B2 using one half standard deviation shocks	86	90	95	88	86	83	74	56
B4. One-time 30 percent real depreciation in 2006	86	103	99	93	91	89	79	63
B5. 10 percent of GDP increase in other debt-creating flows in 2006	85	89	84	78	75	72	60	38
NPV of Debt-to-Revenue Ratio 2/								
Program Scenario	175	172	176	157	155	148	122	68
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	172	182	199	193	206	211	243	304
A2. Permanently lower GDP growth 1/	173	175	182	167	172	171	189	292
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	175	188	215	202	213	215	243	298
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	173	186	209	188	189	183	167	126
B3. Combination of B1-B2 using one half standard deviation shocks	173	190	216	195	197	191	174	133
B4. One-time 30 percent real depreciation in 2006	173	218	226	206	209	204	187	149
B5. 10 percent of GDP increase in other debt-creating flows in 2006	172	189	193	173	173	166	142	89
Debt Service-to-Revenue Ratio 2/								
Program Scenario	10	10	9	9	11	9	11	16
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	10	10	12	15	21	20	27	54
A2. Permanently lower GDP growth 1/	10	11	10	11	14	13	21	51
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	10	11	13	15	19	19	27	52
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	10	11	16	19	17	14	16	26
B3. Combination of B1-B2 using one half standard deviation shocks	10	11	15	18	17	15	17	28
B4. One-time 30 percent real depreciation in 2006	10	12	12	12	15	14	17	27
B5. 10 percent of GDP increase in other debt-creating flows in 2006	10	10	17	13	14	12	13	20

Sources: Country authorities; and Fund staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.

Table A2a. Grenada: External Debt Sustainability Framework, Program Scenario, 2002–2025 1/
(In percent of GDP, unless otherwise indicated)

	Actual		Historical	Standard	Est.	Projections								
	2003	2004	Average 6/	Deviation 6/	2005	2006	2007	2008	2009	2010	2005–10 Average	2015	2025	2011–25 Average
External government debt (nominal) 1/	**80.2**	**95.1**			**84.3**	**81.3**	**77.4**	**73.2**	**69.1**	**65.2**		**48.6**	**26.4**	
Change in external debt	-3.4	15.0			-10.8	-3.0	-3.9	-4.1	-4.1	-3.9		-2.8	-2.6	
Identified net debt-creating flows	6.3	2.4			22.0	9.1	4.9	3.8	2.0	0.8		-4.5	-3.9	
Noninterest current account deficit	**26.9**	**5.3**	**19.0**	**7.2**	**32.8**	**28.7**	**24.0**	**22.6**	**21.6**	**19.6**		**11.2**	**9.9**	12.1
Deficit in balance of goods and services	30.2	31.4			48.0	39.8	34.2	33.6	30.3	27.2		19.6	19.4	
Exports of goods and services	40.7	44.1			35.3	33.8	37.4	37.3	39.7	42.3		45.6	45.8	
Of which : goods	10.3	8.5			6.6	6.1	5.2	5.7	6.1	6.6		6.6	6.6	
Imports of goods and services	70.9	75.6			83.2	73.6	71.6	70.8	70.0	69.5		65.2	65.2	
Net current transfers (negative = inflow)	-8.2	-27.8	-9.0	7.7	-19.5	-15.4	-13.8	-14.0	-12.6	-10.6		-10.8	-11.0	-10.8
Other current account flows (negative = net inflow)	4.9	1.7			4.3	4.3	3.6	3.1	3.8	3.0		2.4	1.5	
Net FDI (negative = inflow)	**-20.1**	**-12.4**	**-13.2**	**3.4**	**-10.5**	**-16.0**	**-17.1**	**-18.0**	**-18.3**	**-18.3**		**-16.4**	**-14.9**	-16.1
Endogenous debt dynamics 2/	**-0.4**	**9.5**			**-0.3**	**-3.6**	**-1.9**	**-0.9**	**-1.2**	**-0.4**		**0.7**	**1.1**	
Contribution from nominal interest rate	6.4	8.2			1.0	1.4	1.8	2.0	1.5	2.2		2.7	2.2	
Contribution from real GDP growth	-4.5	2.5			-1.3	-5.0	-3.8	-2.9	-2.7	-2.6		-1.9	-1.1	
Contribution from price and exchange rate changes	-2.4	-1.2			…	…	…	…	…	…		…	…	
Residual (3–4) 3/	**-9.7**	**12.6**			**-32.9**	**-12.1**	**-8.8**	**-7.9**	**-6.2**	**-4.8**		**1.7**	**1.4**	
Of which : exceptional financing	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt	...	48.4			48.3	48.8	48.4	47.9	47.5	46.5		39.1	22.1	
In percent of exports	...	109.7			136.8	144.3	129.5	128.6	119.6	110.0		85.9	48.2	
NPV of PPG external debt 4/	**...**	**48.4**			**48.3**	**48.8**	**48.4**	**47.9**	**47.5**	**46.5**		**39.1**	**22.1**	
In percent of exports	...	109.7			136.8	144.3	129.5	128.6	119.6	110.0		85.9	48.2	
Debt service-to-exports ratio (in percent)	49.8	64.3			7.5	9.0	8.8	10.0	8.1	8.4		10.8	17.0	
PPG debt service-to-exports ratio (in percent)	49.8	64.3			7.5	9.0	8.8	10.0	8.1	8.4		10.8	17.0	
Total gross financing need (billions of U.S. dollars)	120.0	93.2			118.2	81.6	56.6	49.9	40.8	32.2		-2.6	44.3	
Noninterest current account deficit that stabilizes debt ratio	30.2	-9.7			43.6	31.7	27.9	26.8	25.7	23.5		14.0	12.4	
Key macroeconomic assumptions														
Real GDP growth (in percent)	5.8	-3.0	3.2	4.3	1.5	6.5	5.0	4.0	4.0	4.0	4.2	4.0	4.0	4.0
GDP deflator in US dollar terms (change in percent)	2.9	1.5	1.5	1.0	6.9	2.9	2.5	2.6	2.0	2.0	3.2	2.0	2.0	2.0
Effective interest rate (percent) 5/	8.3	10.1	10.2	2.1	1.2	1.8	2.4	2.8	2.2	3.3	2.3	5.5	8.2	5.8
Growth of exports of goods and services (U.S. dollar terms, in p	5.0	6.8	6.0	17.0	-13.3	5.0	19.0	6.5	13.0	12.8	7.2	7.7	6.1	6.7
Growth of imports of goods and services (U.S. dollar terms, in p	15.3	5.0	6.9	10.5	19.5	-3.1	4.6	5.7	4.8	5.3	6.1	4.8	6.1	5.6
Grant element of new public sector borrowing (in percent)	...	...	...	...	39.0	42.9	39.9	38.0	37.2	36.1	38.8	24.9	12.4	19.9
Memorandym item:														
Nominal GDP (billions of US dollars)	443.7	436.8			474.0	519.3	558.9	596.5	632.5	670.9		902.0	1,630.7	

Source: Fund staff simulations.

1/ Data on private sector external debt is not available.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Based on a discout rate of 9 percent and assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table A2b. Grenada: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2005-25
(In percent)

	Est.	Projections						
	2005	2006	2007	2008	2009	2010	**2015**	2025
NPV of Debt-to-GDP Ratio								
Program Scenario	48	49	48	48	48	46	**39**	22
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	48	46	46	47	50	54	**81**	121
A2. New public sector loans on less favorable terms in 2006-25 2/	48	50	50	50	50	49	**44**	33
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	48	53	55	55	54	53	**45**	25
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	48	53	63	62	61	59	**48**	24
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2006-07	48	50	51	50	50	49	**41**	23
B4. Net nondebt creating flows at historical average minus one standard deviation in 2006-07 4/	48	65	80	78	76	73	**57**	26
B5. Combination of B1–B4 using one-half standard deviation shocks	48	63	82	81	79	76	**60**	28
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	48	69	68	68	67	66	**55**	31
NPV of Debt-to-Exports Ratio								
Program Scenario	137	144	129	129	120	110	**86**	48
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	137	135	122	127	126	127	**178**	264
A2. New public sector loans on less favorable terms in 2006-25 2/	137	147	134	134	125	116	**97**	72
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	137	144	129	129	120	110	**86**	48
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	137	184	267	264	243	222	**166**	82
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2006-07	137	144	129	129	120	110	**86**	48
B4. Net nondebt creating flows at historical average minus one standard deviation in 2006-07 4/	137	191	213	210	191	173	**125**	56
B5. Combination of B1–B4 using one-half standard deviation shocks	137	188	256	253	231	210	**154**	72
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	137	144	129	129	120	110	**86**	48
Debt Service Ratio								
Program Scenario	8	9	9	10	8	8	**11**	17
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	8	9	9	7	7	8	**18**	49
A2. New public sector loans on less favorable terms in 2006-25 2/	8	9	8	8	8	8	**9**	18
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	8	9	9	9	8	8	**10**	15
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	8	11	15	19	20	20	**21**	25
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2006-07	8	9	9	9	8	8	**10**	15
B4. Net nondebt creating flows at historical average minus one standard deviation in 2006-07 4/	8	9	11	17	18	18	**17**	17
B5. Combination of B1–B4 using one-half standard deviation shocks	8	10	14	19	21	20	**20**	22
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	8	9	9	9	8	8	**10**	15
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	22	22	22	22	22	22	**22**	22

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

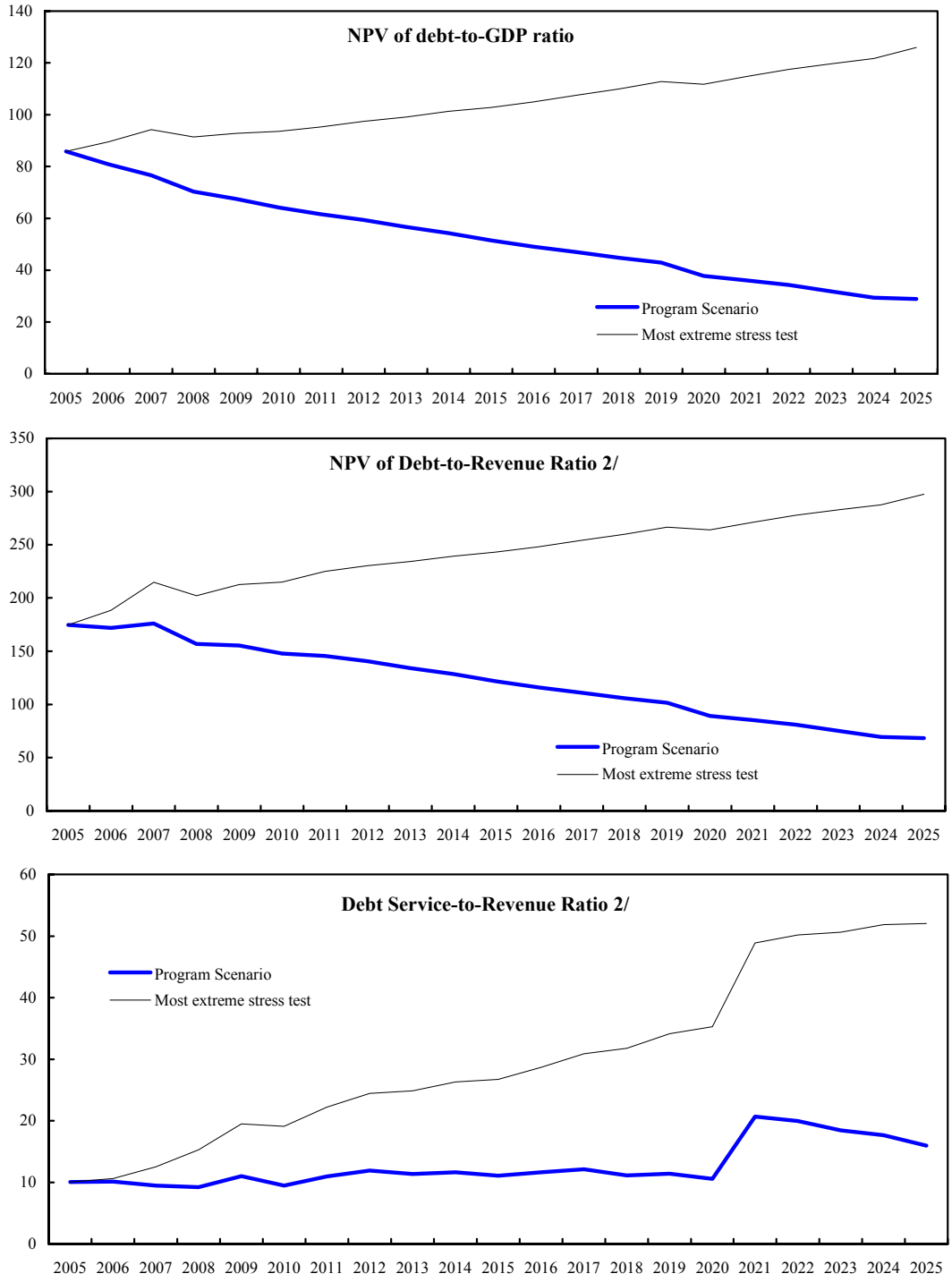
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

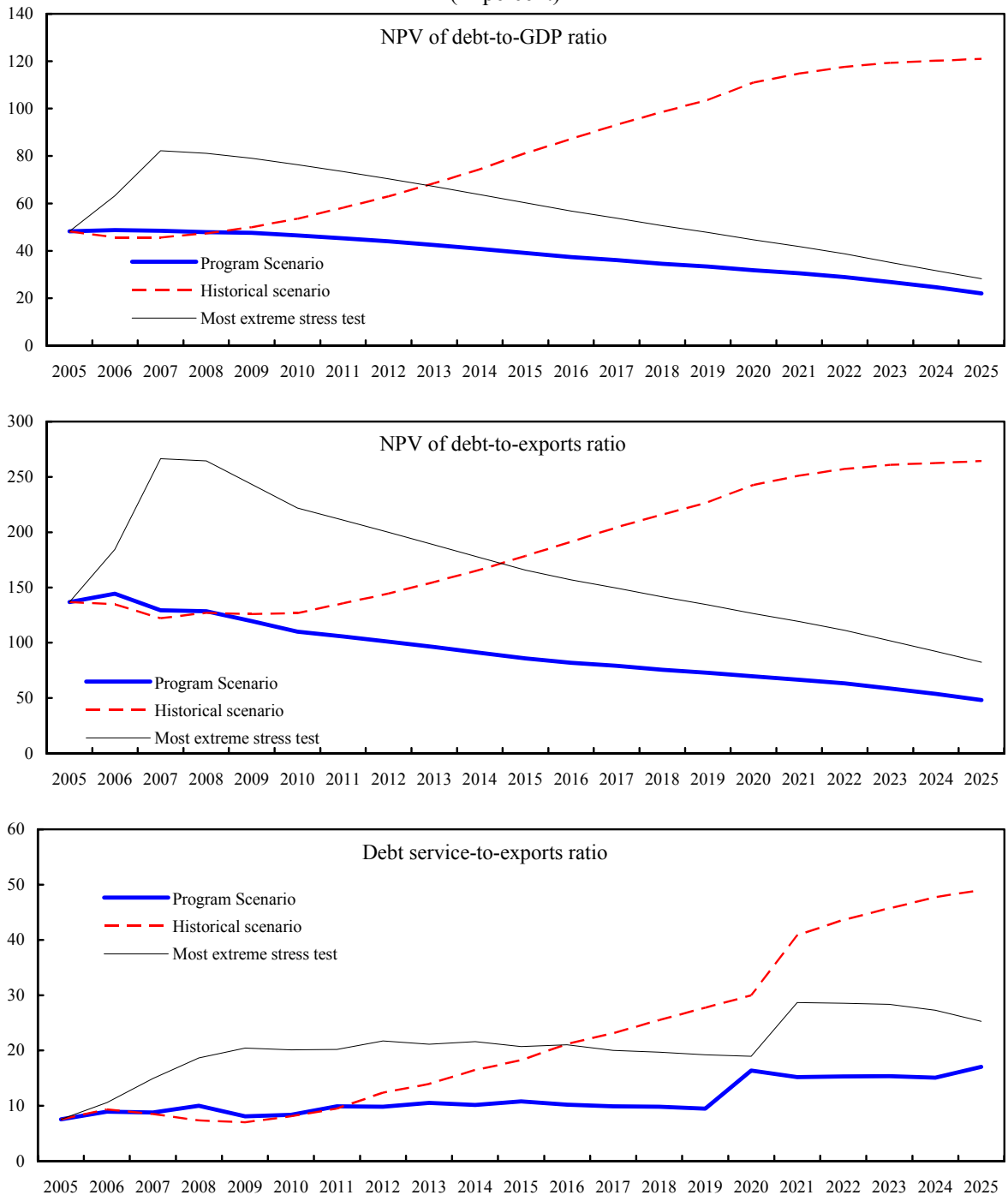
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 1.Grenada: Indicators of Public Debt Under Alternative Scenarios, 2005-2025 1/



Source: Fund staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2015.
2/ Revenue including grants.

Figure 2. Grenada: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2005–2025
(In percent)



Source: Fund staff projections and simulations.